



CY 104 – Introduction to Cyber Security

lab # 03

Networking Fundamentals, Security Commands and Linux Basics

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TASKS

The lab is to practice basic system and network inspection commands Kali Linux. For every task, the commands I typed are recorded with a screenshot of the result.

Task 1: Identify Network Configuration

Identifying the IP address, subnet mask, and default gateway of my system.

```
C:\Users\2604277>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Alpha26
Primary Dns Suffix . . . . . : austd.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : austd.com

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-04
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ef82:4dcc:b09f:558f%4(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 420085799
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-07-B3-19-B0-7B-25-13-89-FD
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 4:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (5) I219-V
Physical Address. . . . . : D8-9E-F3-3E-0F-67
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::b20d:a28e:5994:f85f%12(Preferred)
IPv4 Address. . . . . : 172.30.1.26(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 172.30.0.196
DHCPv6 IAID . . . . . : 475044428
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-07-B3-19-B0-7B-25-13-89-FD
DNS Servers . . . . . : 172.30.0.201
                        172.30.0.202
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
```

IPv4 Address	172.30.1.26
Subnet Mask	255.255.0.0
Default Gateway	172.30.0.196
DNS Server(s)	172.30.0.201, 172.30.0.202

The output provided important network information (IP address, subnet mask, gateway, and DNS configuration). This information is helpful in understand the device's network environment for an authorized administrator.

Task 2: Connectivity and Latency with PING

Using ping to check connectivity:

```
C:\Users\2604277>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=39ms TTL=115
Reply from 8.8.8.8: bytes=32 time=34ms TTL=115
Reply from 8.8.8.8: bytes=32 time=35ms TTL=115
Reply from 8.8.8.8: bytes=32 time=34ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 34ms, Maximum = 39ms, Average = 35ms
```

```
C:\Users\2604277>ping au.edu.pk

Pinging au.edu.pk [104.21.79.203] with 32 bytes of data:
Reply from 104.21.79.203: bytes=32 time=20ms TTL=56
Reply from 104.21.79.203: bytes=32 time=20ms TTL=56
Reply from 104.21.79.203: bytes=32 time=22ms TTL=56
Reply from 104.21.79.203: bytes=32 time=20ms TTL=56

Ping statistics for 104.21.79.203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 22ms, Average = 20ms
```

```
C:\Users\2604277>ping nasa.gov

Pinging nasa.gov [192.0.66.108] with 32 bytes of data:
Reply from 192.0.66.108: bytes=32 time=99ms TTL=56
Reply from 192.0.66.108: bytes=32 time=98ms TTL=56
Reply from 192.0.66.108: bytes=32 time=99ms TTL=56
Reply from 192.0.66.108: bytes=32 time=98ms TTL=56

Ping statistics for 192.0.66.108:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 98ms, Maximum = 99ms, Average = 98ms
```

Site	Status	Average Latency	Observation
Google (google.com)	Reached	35ms	Good connectivity with normal response time.
Air university (au.edu.pk)	Reached	20ms	Fastest response time among the tested sites.
NASA (nasa.gov)	Reached	98ms	Highest latency, likely due to the longer network path.

Ping responses showed whether the destination was reachable and provided the approximate network latency. Higher response times can indicate a longer network path, congestion, or other network condition.

Task 3: Trace a Network Path

Performing tracert to different international sites and documenting the number of hops:

```
C:\Users\2604277>tracert bbc.co.uk

Tracing route to bbc.co.uk [151.101.128.81]
over a maximum of 30 hops:

  1    1 ms    <1 ms    <1 ms    172.30.0.196
  2    2 ms     1 ms     1 ms    au.edu.pk [111.68.96.97]
  3    3 ms     3 ms     2 ms    172.31.252.49
  4    2 ms     2 ms     1 ms    172.31.240.8
  5    1 ms     1 ms     1 ms    117.20.23.233
  6   21 ms    20 ms    21 ms    110.93.255.186
  7  126 ms   104 ms   100 ms   110.93.252.136
  8  158 ms   174 ms    *      195.229.27.221
  9  445 ms   462 ms   466 ms   5.195.70.162
 10  530 ms   558 ms   590 ms   140.248.64.46
 11   63 ms    41 ms    45 ms   151.101.128.81

Trace complete.
```

```
C:\Users\2604277>tracert yahoo.co.jp

Tracing route to yahoo.co.jp [124.83.187.124]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms    172.30.0.196
  2    2 ms     1 ms     1 ms    au.edu.pk [111.68.96.97]
  3    6 ms    12 ms     8 ms    172.31.252.49
  4    4 ms     3 ms     6 ms    172.31.240.8
  5    2 ms     4 ms     2 ms    117.20.23.233
  6   21 ms    20 ms    26 ms    110.93.253.182
  7   23 ms    20 ms    23 ms    110.93.252.246
  8  140 ms   144 ms   141 ms    be5685.agr21.fra06.atlas.cogentco.com [149.14.211.241]
  9  355 ms   144 ms   142 ms    be2844.rcr22.fra06.atlas.cogentco.com [130.117.0.29]
 10  142 ms   142 ms   141 ms    be7948.ccr42.fra05.atlas.cogentco.com [154.54.72.126]
 11  146 ms   145 ms   148 ms    be2950.ccr42.ams03.atlas.cogentco.com [154.54.72.41]
 12  151 ms   154 ms   153 ms    be2183.ccr22.lpl01.atlas.cogentco.com [154.54.58.69]
 13  228 ms   224 ms   224 ms    be3043.ccr22.ymq01.atlas.cogentco.com [154.54.44.166]
 14  230 ms   234 ms   229 ms    be3260.ccr32.yyz02.atlas.cogentco.com [154.54.42.89]
 15  285 ms   285 ms   286 ms    be4941.ccr82.sea08.atlas.cogentco.com [154.54.94.73]
 16  285 ms   281 ms   290 ms    be9343.ccr22.sea02.atlas.cogentco.com [154.54.160.242]
 17  291 ms   307 ms   307 ms    te0-4-0-27.ccr21.sea02.atlas.cogentco.com [38.104.126.85]
 18  404 ms   404 ms   405 ms    osk004bb01.IIJ.Net [58.138.88.197]
 19  414 ms   412 ms   413 ms    osk004ip57.IIJ.Net [58.138.106.166]
 20  307 ms   305 ms   304 ms    210.130.135.130
 21    *        *        *      Request timed out.
 22  354 ms   349 ms   365 ms    124.83.187.124

Trace complete.
```

Destination	No. of Hops	Observation
bbc.co.uk	11	Latency increased at some hops, with one hop showing no response, but the destination was successfully reached.
yahoo.co.jp	22	The route reached Japan in 22 hops. The path involved international networks and showed higher latency, with one hop timing out.

The tracert shows that traffic towards two destinations is following different network paths. Yahoo Japan had required more hops and had higher latency, while BBC had reached in lesser hops. Some routers did not respond to traceroute.

Task 4: Inspect Listening Ports and Connections

Runing netstat on my system and identifying listening ports and associated process IDs.

TCP	[::]:21	[::]:0	LISTENING	4348
TCP	[::]:80	[::]:0	LISTENING	4136
TCP	[::]:135	[::]:0	LISTENING	1384
TCP	[::]:443	[::]:0	LISTENING	4136
TCP	[::]:445	[::]:0	LISTENING	4
TCP	[::]:623	[::]:0	LISTENING	4772
TCP	[::]:3306	[::]:0	LISTENING	4824
TCP	[::]:4369	[::]:0	LISTENING	6448
TCP	[::]:5150	[::]:0	LISTENING	6656
TCP	[::]:5500	[::]:0	LISTENING	8464
TCP	[::]:7680	[::]:0	LISTENING	12460
TCP	[::]:8030	[::]:0	LISTENING	4
TCP	[::]:8080	[::]:0	LISTENING	7304
TCP	[::]:8999	[::]:0	LISTENING	4
TCP	[::]:16992	[::]:0	LISTENING	4772
TCP	[::]:25734	[::]:0	LISTENING	6648
TCP	[::]:49664	[::]:0	LISTENING	1096
TCP	[::]:49665	[::]:0	LISTENING	404
TCP	[::]:49666	[::]:0	LISTENING	1940
TCP	[::]:49667	[::]:0	LISTENING	2532
TCP	[::]:49668	[::]:0	LISTENING	3976
TCP	[::]:49669	[::]:0	LISTENING	1096
TCP	[::]:49761	[::]:0	LISTENING	7008
TCP	[::]:50062	[::]:0	LISTENING	1068
TCP	[::]:50064	[::]:0	LISTENING	11672
TCP	[::]:50189	[::]:0	LISTENING	8592

Local Address / Port	State	PID	Service / Process	Observation
[::]:21	LISTENING	4348	FTP service	Port 21 is listening for FTP connections.
[::]:80	LISTENING	4136	HTTP service	Port 80 is listening for HTTP web connections.
[::]:135	LISTENING	1384	Windows RPC	Port 135 is used by Windows for RPC communication.

The netstat -ano showed many listening but here are three of them for TCP port: 21, 80, and 135. The PIDs can be used with Task Manager to identify the processes using these ports.

Task 5: Process Analysis

```
C:\Users\2604277>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	132 K
Secure System	140	Services	0	32,264 K
Registry	180	Services	0	49,152 K
smss.exe	644	Services	0	28 K
csrss.exe	944	Services	0	2,244 K
wininit.exe	404	Services	0	548 K
services.exe	1068	Services	0	6,516 K
LsaIso.exe	1088	Services	0	2,292 K
lsass.exe	1096	Services	0	18,576 K
svchost.exe	1240	Services	0	42,020 K
fontdrvhost.exe	1300	Services	0	164 K
svchost.exe	1384	Services	0	13,100 K
svchost.exe	1428	Services	0	5,316 K
svchost.exe	1600	Services	0	660 K
svchost.exe	1624	Services	0	44,308 K
svchost.exe	1764	Services	0	2,252 K
svchost.exe	1824	Services	0	944 K
svchost.exe	1820	Services	0	1,948 K
svchost.exe	1836	Services	0	3,588 K
IntelCpHDCPSvc.exe	1864	Services	0	16 K
svchost.exe	1880	Services	0	1,052 K
svchost.exe	1940	Services	0	17,324 K
svchost.exe	1992	Services	0	3,548 K

ai.exe	13908	Console	2	16,380 K
chrome.exe	14776	Console	2	49,132 K
chrome.exe	6484	Console	2	69,888 K
chrome.exe	2704	Console	2	276,712 K
SearchFilterHost.exe	10504	Services	0	10,804 K
SearchFilterHost.exe	3840	Services	0	10,064 K
backgroundTaskHost.exe	18368	Console	2	56,452 K
RuntimeBroker.exe	2916	Console	2	30,104 K
cmd.exe	11124	Console	2	6,348 K
conhost.exe	4416	Console	2	8,852 K
OpenConsole.exe	3212	Console	2	19,020 K
WindowsTerminal.exe	13000	Console	2	112,380 K
SearchFilterHost.exe	14512	Services	0	10,708 K
SnippingTool.exe	15472	Console	2	142,740 K
svchost.exe	13984	Console	2	16,020 K
chrome.exe	10892	Console	2	31,500 K
backgroundTaskHost.exe	16260	Console	2	5,732 K
PhoneExperienceHost.exe	15336	Console	2	151,056 K
RuntimeBroker.exe	2920	Console	2	29,628 K
backgroundTaskHost.exe	6216	Console	2	15,176 K
SearchFilterHost.exe	13688	Services	0	12,588 K
WmiPrvSE.exe	15172	Services	0	14,676 K
tasklist.exe	4696	Console	2	11,836 K

Process	PID	Reason Selected	Result
tasklist.exe	4696	Command used to view running processes	Running normally
cmd.exe	1112 4	Command Prompt process under use during this lab	Running normally
wazuh-agent.exe	7956	Security monitoring agent running on my system	Running normally

Security analysts examine process names, PIDs, paths, and resource usage to identify unusual or suspicious processes and understand activities of system.

Task 6: Linux File-System Navigation

Current user and directory:

```
(kali㉿kali)-[~]
$ pwd
/home/kali

(kali㉿kali)-[~]
$ whoami
kali
```

Hidden files and permissions:

```
(kali㉿kali)-[~]
$ ls -la
total 140
drwx----- 18 kali kali 4096 Sep 28 03:37 .
drwxr-xr-x  6 root root 4096 Sep 28 03:35 ..
-rw-----  1 kali kali   56 Sep 28 03:36 .bash_history
-rw-r--r--  1 kali kali  220 Sep 28 03:35 .bash_logout
-rw-r--r--  1 kali kali 5578 Sep 28 03:35 .bashrc
-rw-r--r--  1 kali kali 3526 Sep 28 03:35 .bashrc.original
drwxrwxr-x  9 kali kali 4096 Sep 28 03:38 .cache
drwxr-xr-x 13 kali kali 4096 Sep 28 03:37 .config
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Desktop
-rw-r--r--  1 kali kali   35 Sep 28 03:37 .dmrc
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Documents
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Downloads
-rw-r--r--  1 kali kali 11759 Sep 28 03:35 .face
lrwxrwxrwx  1 kali root    5 Sep 28 03:35 .face.icon → .face
drwx-----  3 kali kali 4096 Sep 28 03:37 .gnupg
-rw-----  1 kali kali    0 Sep 28 03:37 .ICEauthority
drwxr-xr-x  3 kali kali 4096 Sep 28 03:35 .java
drwxr-xr-x  5 kali kali 4096 Sep 28 03:37 .local
drwxrwxr-x  5 kali kali 4096 Sep 28 03:38 .mozilla
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Music
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Pictures
-rw-r--r--  1 kali kali  807 Sep 28 03:35 .profile
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Projects
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Public
drwx-----  3 kali kali 4096 Sep 28 03:37 .ssh
-rw-r--r--  1 kali kali    0 Sep 28 03:36 .sudo_as_admin_successful
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Templates
drwxr-xr-x  2 kali kali 4096 Sep 28 03:37 Videos
-rw-----  1 kali kali   49 Sep 28 03:37 .Xauthority
-rw-----  1 kali kali 7331 Sep 28 03:37 .xsession-errors
-rw-r--r--  1 kali kali  336 Sep 28 03:35 .zprofile
-rw-r--r--  1 kali kali 10882 Sep 28 03:35 .zshrc

(kali㉿kali)-[~]
$ la
.bash_history      Documents          .mozilla          Templates
.bash_logout      Downloads         Music            Videos
.bashrc            .face            Pictures          .Xauthority
.bashrc.original  .face.icon       profile          .xsession-errors
.cache             .gnupg           Projects         .zprofile
.config            .ICEauthority    Public           .zshrc
Desktop            .java            .ssh
.dmrc              .local           .sudo_as_admin_successful
```

Files with size:

```
(kali㉿kali)-[~]
$ la -ls
total 132
4 -rw----- 1 kali kali 56 Sep 28 03:36 .bash_history
4 -rw-r--r-- 1 kali kali 220 Sep 28 03:35 .bash_logout
8 -rw-r--r-- 1 kali kali 5578 Sep 28 03:35 .bashrc
4 -rw-r--r-- 1 kali kali 3526 Sep 28 03:35 .bashrc.original
4 drwxrwxr-x 9 kali kali 4096 Sep 28 03:38 .cache
4 drwxr-xr-x 13 kali kali 4096 Sep 28 03:37 .config
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Desktop
4 -rw-r--r-- 1 kali kali 35 Sep 28 03:37 .dmrc
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Documents
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Downloads
12 -rw-r--r-- 1 kali kali 11759 Sep 28 03:35 .face
0 lrwxrwxrwx 1 kali root 5 Sep 28 03:35 .face.icon → .face
4 drwx----- 3 kali kali 4096 Sep 28 03:37 .gnupg
0 -rw----- 1 kali kali 0 Sep 28 03:37 .ICEauthority
4 drwxr-xr-x 3 kali kali 4096 Sep 28 03:35 .java
4 drwxr-xr-x 5 kali kali 4096 Sep 28 03:37 .local
4 drwxrwxr-x 5 kali kali 4096 Sep 28 03:38 .mozilla
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Music
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Pictures
4 -rw-r--r-- 1 kali kali 807 Sep 28 03:35 .profile
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Projects
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Public
4 drwx----- 3 kali kali 4096 Sep 28 03:37 .ssh
0 -rw-r--r-- 1 kali kali 0 Sep 28 03:36 .sudo_as_admin_successful
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Templates
4 drwxr-xr-x 2 kali kali 4096 Sep 28 03:37 Videos
4 -rw----- 1 kali kali 49 Sep 28 03:37 .Xauthority
8 -rw----- 1 kali kali 7331 Sep 28 03:37 .xsession-errors
4 -rw-r--r-- 1 kali kali 336 Sep 28 03:35 .zprofile
12 -rw-r--r-- 1 kali kali 10882 Sep 28 03:35 .zshrc
```

Creating a folder and file and inserting some text in it.

```
(kali㉿kali)-[~]
$ mkdir cyber-lab

(kali㉿kali)-[~]
$ cd cyber-lab/

(kali㉿kali)-[~/cyber-lab]
$ touch evidence.txt

(kali㉿kali)-[~/cyber-lab]
$ echo "A evidence in it." > evidence.txt
```

Back up and renaming:

```
(kali㉿kali)-[~/cyber-lab]
$ cp evidence.txt evidence_backup.txt

(kali㉿kali)-[~/cyber-lab]
$ mv evidence.txt investigation.txt
```

Files:

```
(kali㉿kali)-[~/cyber-lab]
$ ls -la
total 16
drwxrwxr-x  2 kali kali 4096 Sep 28 03:41 .
drwx----- 19 kali kali 4096 Sep 28 03:39 ..
-rw-rw-r--  1 kali kali   18 Sep 28 03:41 evidence_backup.txt
-rw-rw-r--  1 kali kali   18 Sep 28 03:40 investigation.txt
```

OS deatils:

```
(kali㉿kali)-[~/cyber-lab]
$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```

System configuration files:

```
(kali㉿kali)-[~/cyber-lab]
└─$ ls /etc
adduser.conf          ipsec.d              rc0.d
alternatives          ipsec.secrets        rc1.d
apache2               issue               rc2.d
apparmor              issue.net            rc3.d
apparmor.d            java-25-openjdk      rc4.d
apt                   john                 rc5.d
arp-scan              kali-menu            rc6.d
avahi                 kernel               rcS.d
bash.bashrc           keyutils             reader.conf.d
bash_completion       kismet              rearj.cfg
bash_completion.d     ldap                 redis
bindresvport.blacklist ld.so.cache          redsocks.conf
binfmt.d              ld.so.conf           request-key.conf
bluetooth             ld.so.conf.d         request-key.d
ca-certificates       legion.conf          resolv.conf
ca-certificates.conf  libao.conf           responder
chatscripts           libaudit.conf        rmt
chromium              libblockdev          rpc
chromium.d            libccid_Info.plist  runit
cifs-utils            libnl-3              samba
cloud                 libpaper.d           sane.d
colord                lightdm              scalpel
console-setup         lighttpd             screenrc
cracklib              locale.conf          sddm.conf.d
credstore             locale.gen           searchsploit_rc
credstore.encrypted  localtime            security
cron.d                logcheck             selinux
cron.daily            login.defs           sensors3.conf
cron.hourly           logrotate.conf       sensors.d
cron.monthly          logrotate.d          services
```

Log files:

```
(kali㉿kali)-[~/cyber-lab]
└─$ ls /var/log
alternatives.log      private              vmware-vmtoolsd-ahmedsajjad.log
apache2              redis               vmware-vmtoolsd-alihasan66879.log
apt                  runit               vmware-vmtoolsd-kali.log
boot.log             samba               vmware-vmtoolsd-root.log
boot.log.1           speech-dispatcher   vmware-vmtoolsd-tempadmin.log
btmtp                stunnel             vmware-vmusr-ahmedsajjad.log
dpkg.log             sysstat             vmware-vmusr-alihasan66879.log
fontconfig.log       vmware-network.1.log vmware-vmusr-kali.log
gvm                  vmware-network.2.log vmware-vmusr-tempadmin.log
inetsim              vmware-network.3.log wtmp
journal              vmware-network.4.log wtmp.db
lastlog              vmware-network.5.log Xorg.0.log
lightdm              vmware-network.6.log Xorg.0.log.old
macchanger.log       vmware-network.7.log Xorg.1.log
mosquitto            vmware-network.8.log Xorg.1.log.old
mysql                vmware-network.9.log Xorg.2.log
nginx                vmware-network.log  Xorg.2.log.old
notus-scanner        vmware-vmtoolsd-1.log
openvpn              vmware-vmtoolsd-2.log
postgresql            vmware-vmtoolsd-3.log

(kali㉿kali)-[~/cyber-lab]
```

1- Difference between /home, /etc, and /var/log:

/home - user files

/etc - system configuration files

/var/log - system & application log files.

2- Difference between ls and ls -la:

ls - normal files & directories

ls -la - hidden files & detailed file information.

3- Why are hidden files useful during a security investigation?

Hidden files may contain config information or files that provide useful information during an investigation.

4- Why inspect /var/log after a security incident?

Logs contain information about activity, authentication attempts, errors, and other events that helps knowing what happened.

5- Why should original evidence files not be modified unnecessarily?

Original evidence should be preserved so that its integrity is maintained and the information remains same for investigation.

Wireshark

In Kali Linux, I have started wireshark capture, and generated ping request to google. Here is the result I found on wirehsark:

The image shows a Wireshark capture window with the filter `ip.addr == 8.8.8.8`. The packet list shows a series of ICMP Echo (ping) requests and replies. The packet details pane for packet 504 shows the following structure:

- Frame 504: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface `bnep0`
- Ethernet II, Src: HonHaiPrecis_82:fb:e5 (64:27:37:82:fb:e5), Dst: 08:00:00:00:00:00
- Internet Protocol Version 4, Src: 192.168.44.81, Dst: 8.8.8.8
- Internet Control Message Protocol
 - Type: Echo (ping) request, ID: 0x76f1, Seq: 0x00000000

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```

0000  50 3c ea 2b 63 d4 64 27 37 82 fb e5 08 00 45 00
0010  00 54 c7 45 40 00 40 01 76 5a c0 a8 2c 51 08 00
0020  08 08 08 00 5a 17 76 f1 00 01 82 96 ba 6a 00 00
0030  00 00 28 22 03 00 00 00 00 00 10 11 12 13 14 15
0040  16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
0050  26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
0060  36 37
  
```

The status bar at the bottom indicates: `Packets: 2254 · Displayed: 12 (0.5%) · Dropped: 0 (0.0%) · Profile: Default`

WhoisLookup

To get IP address/domain details we use dnslookup. One example in Linux is whois:

```
anyways @anyways ~ × whois au.edu.pk
# WHOIS .PK Domains (PKNIC)

Domain: au.edu.pk
Status: Domain is Registered

Creation Date: 2002-06-06
Expiry Date: 2028-06-06
Name Server: pam.ns.cloudflare.com
Name Server: vern.ns.cloudflare.com
Name Server:
Name Server:
```